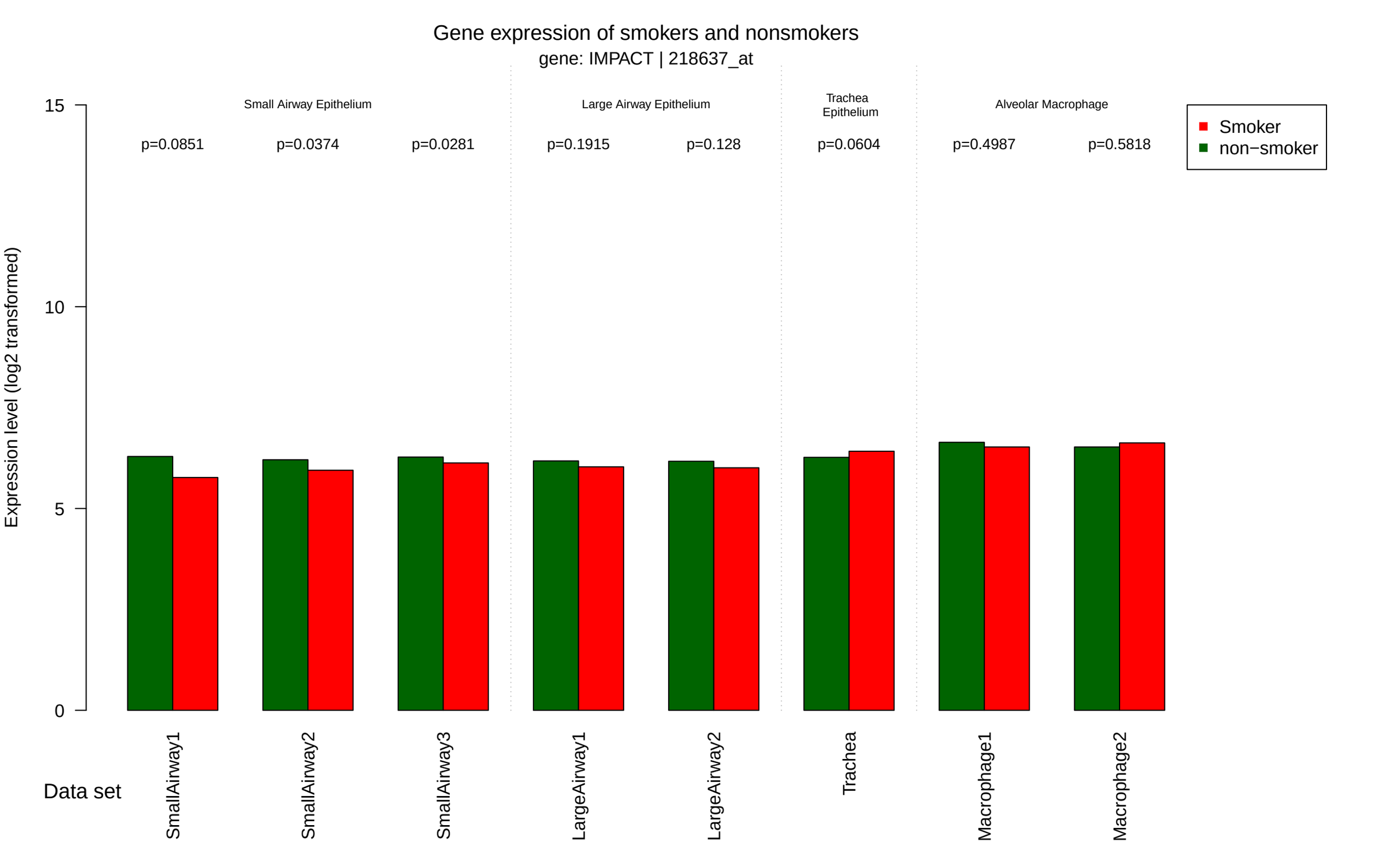
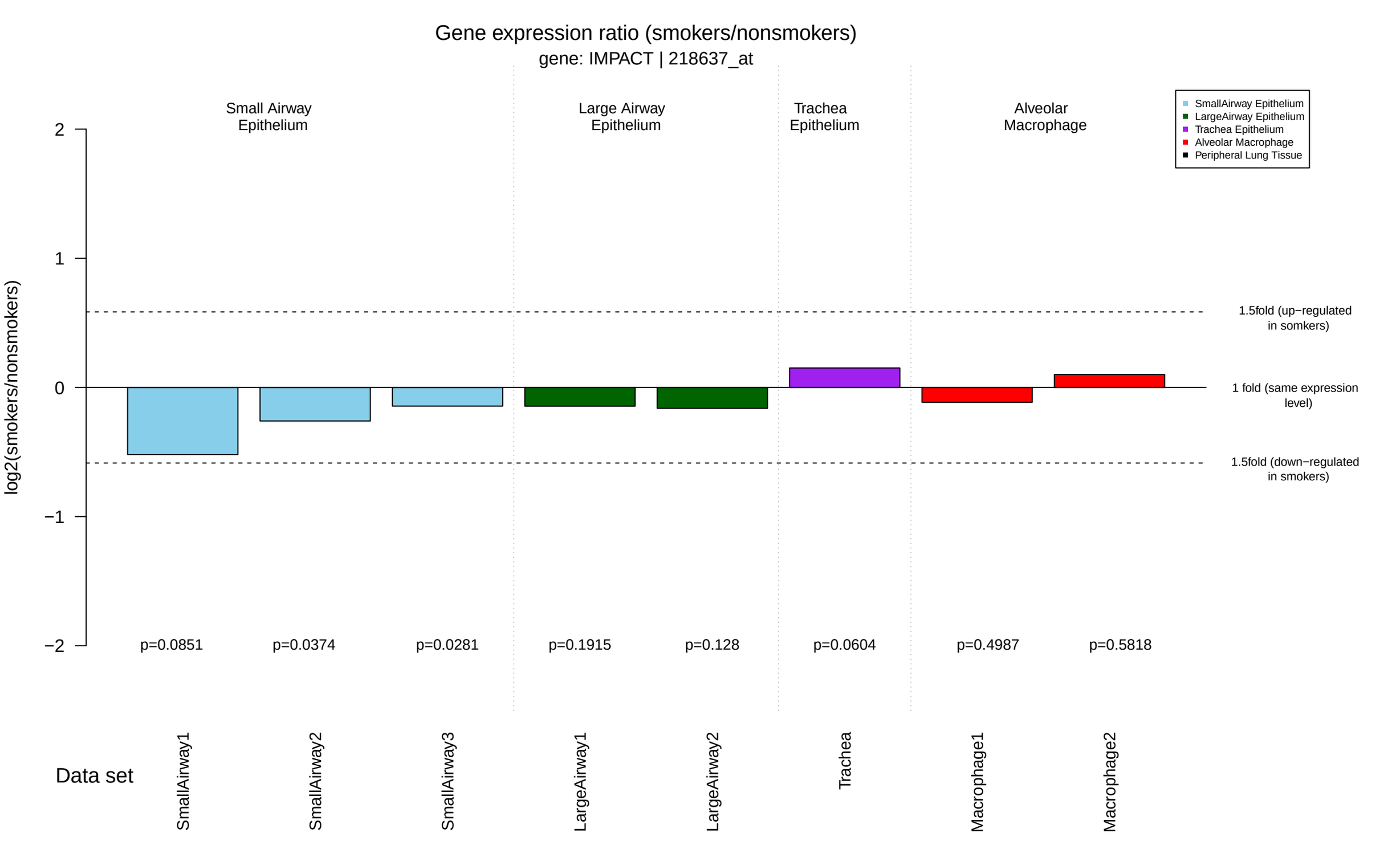
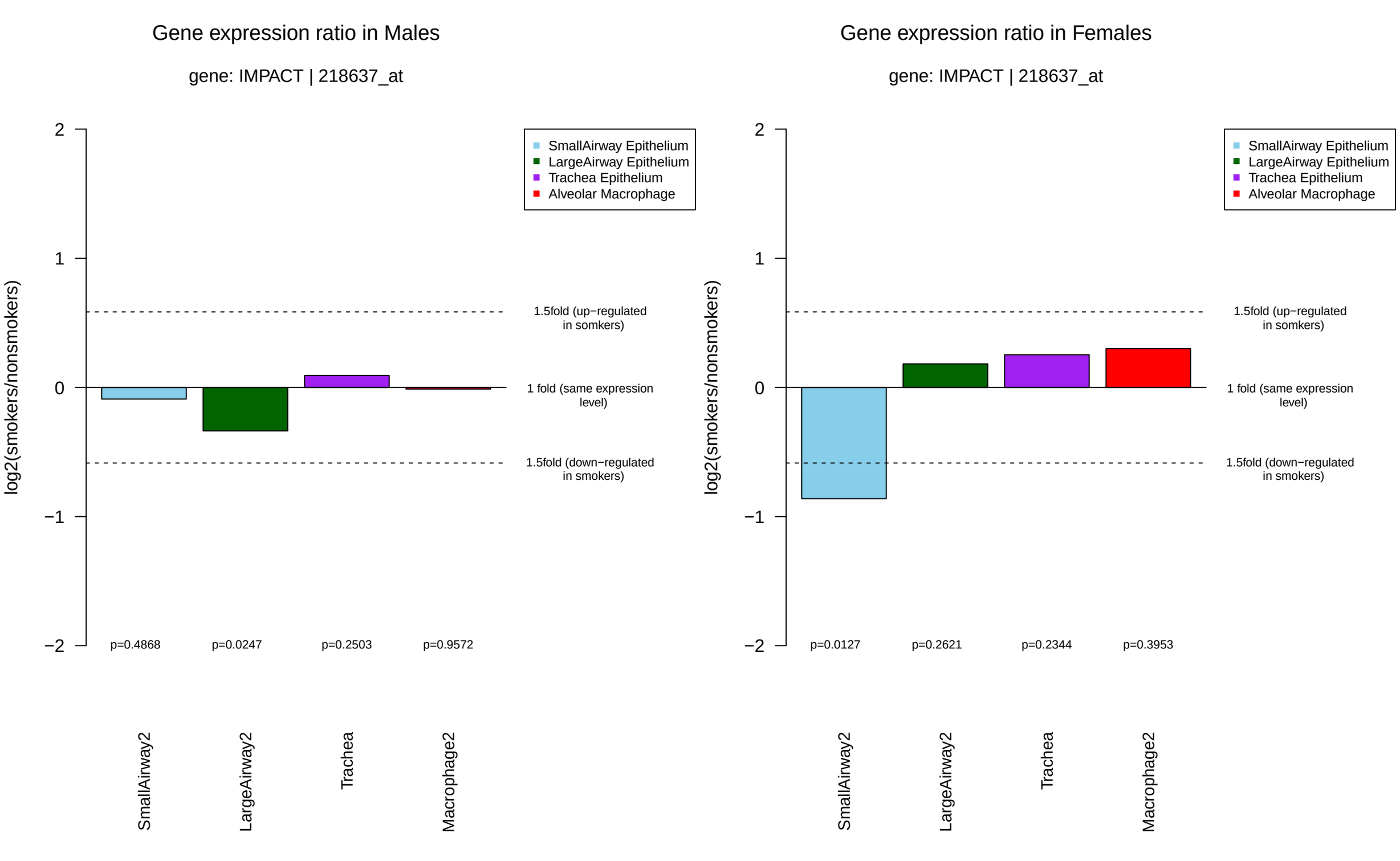
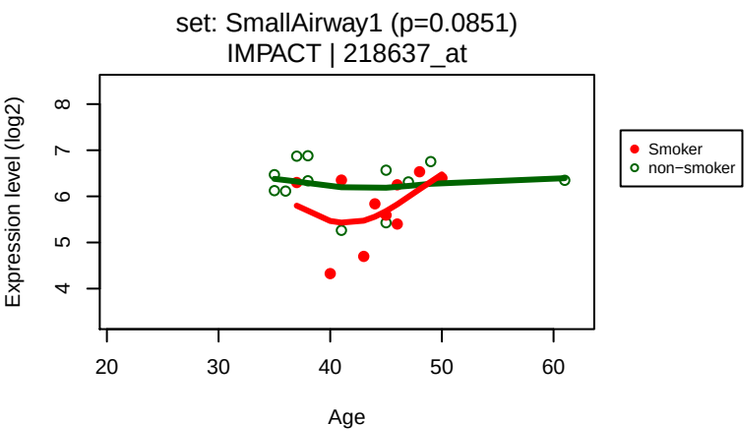
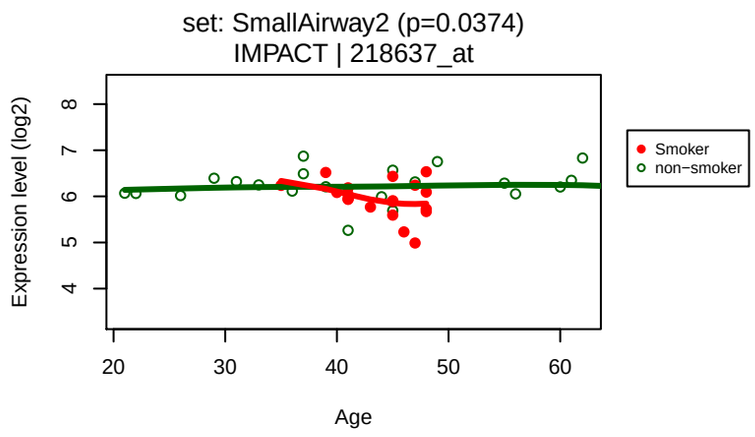
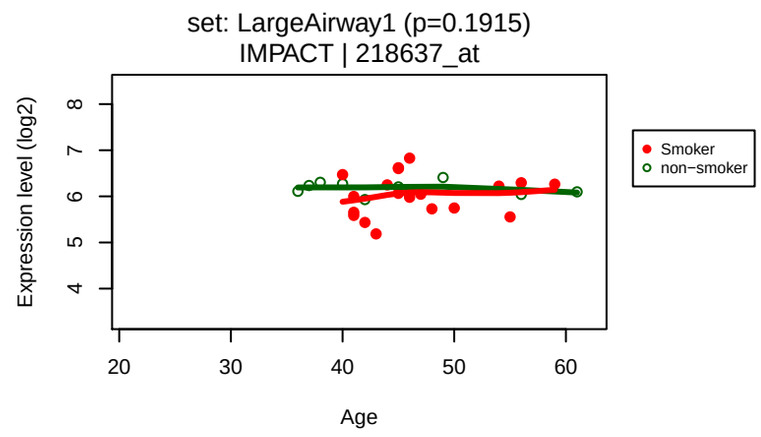

Gene expression of smokers and nonsmokers
gene: IMPACT | 218637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0851
p=0.0374
p=0.0281
p=0.1915
p=0.128
p=0.0604
p=0.4987
p=0.5818
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

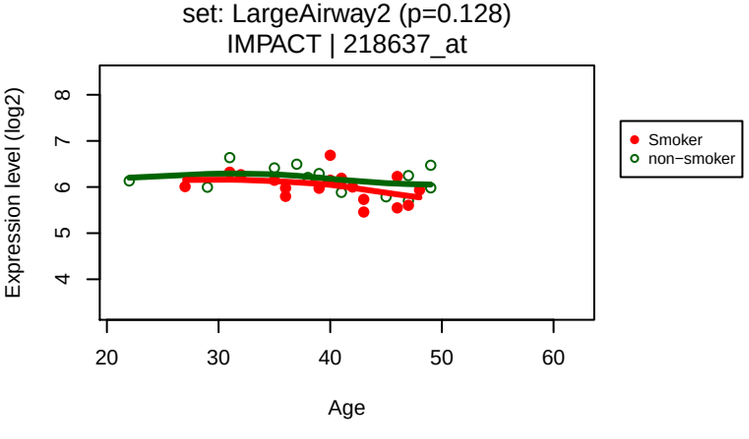

Gene expression ratio (smokers/nonsmokers)
gene: IMPACT | 218637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0851
p=0.0374
p=0.0281
p=0.1915
p=0.128
p=0.0604
p=0.4987
p=0.5818
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

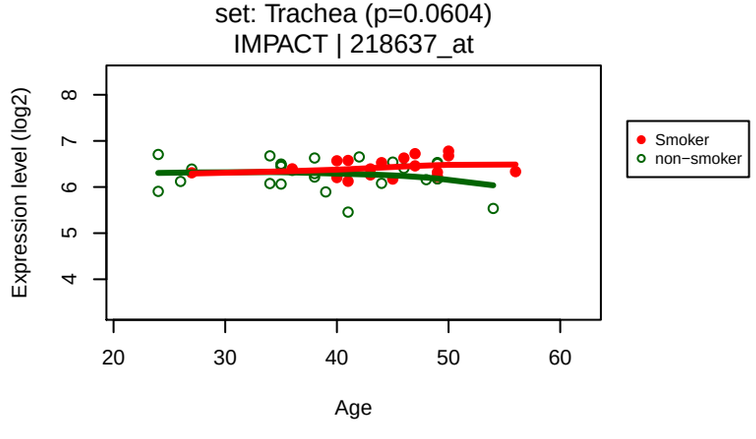

Gene expression ratio in Males
gene: IMPACT | 218637_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4868
p=0.0247
p=0.2503
p=0.9572
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMPACT | 218637_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0127
p=0.2621
p=0.2344
p=0.3953
SmallAirway2
LargeAirway2
Trachea
Macrophage2

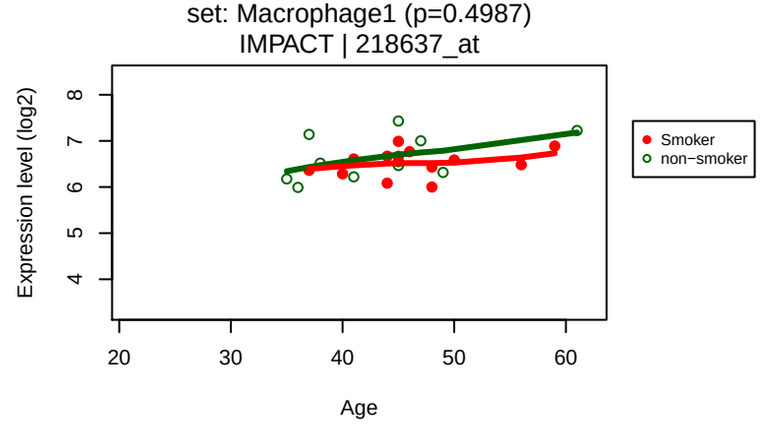

set: SmallAirway1 (p=0.0851)
IMPACT | 218637_at
Expression level (log2)
Age
Smoker
non−smoker

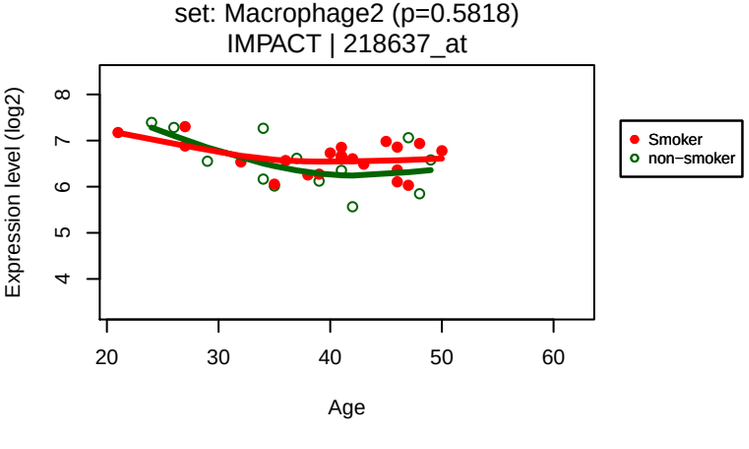

set: SmallAirway2 (p=0.0374)
IMPACT | 218637_at
Expression level (log2)
Age
Smoker
non−smoker

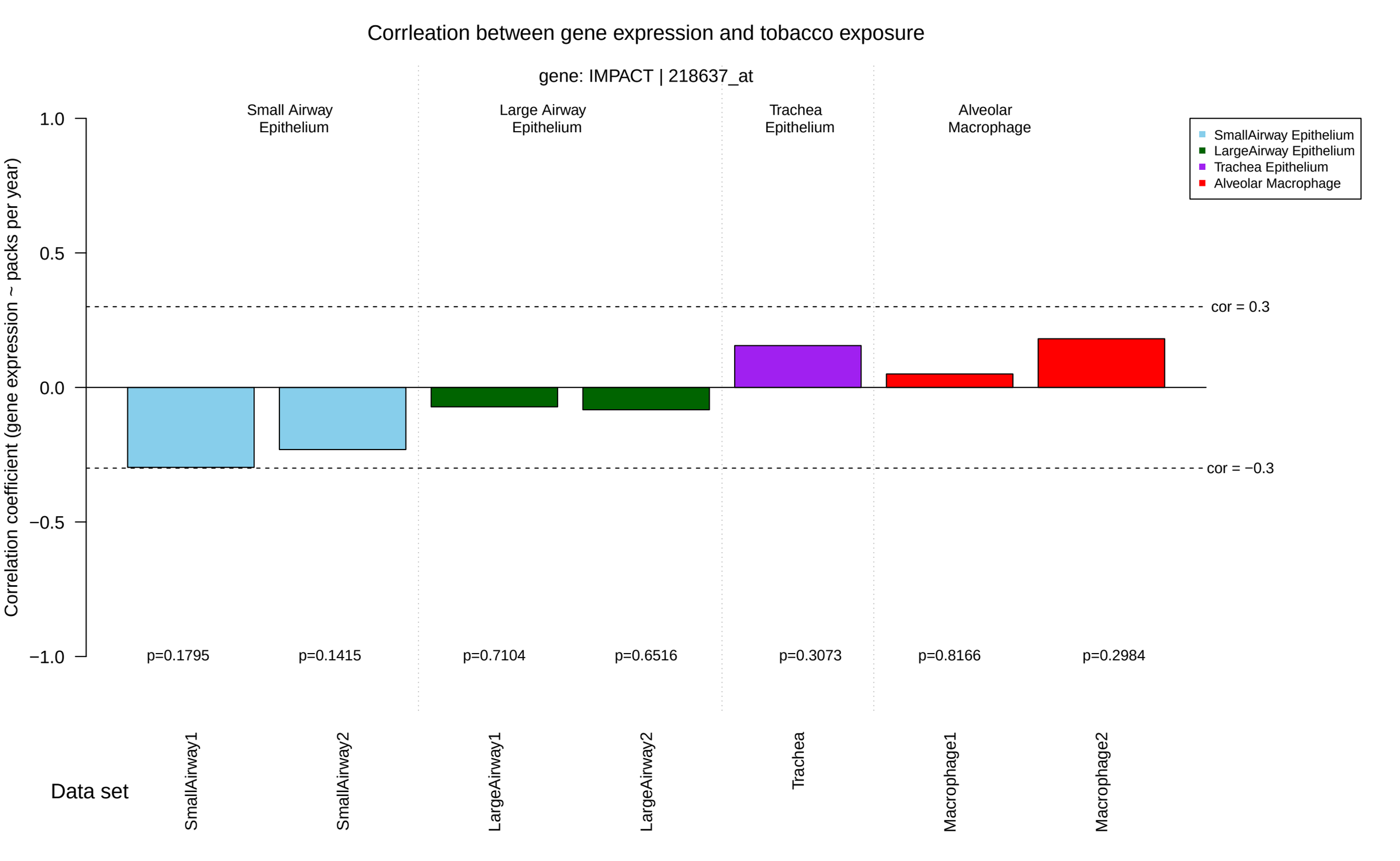

set: LargeAirway1 (p=0.1915)
IMPACT | 218637_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.128)
IMPACT | 218637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0604)
IMPACT | 218637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4987)
IMPACT | 218637_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5818)
IMPACT | 218637_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMPACT | 218637_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1795
p=0.1415
p=0.7104
p=0.6516
p=0.3073
p=0.8166
p=0.2984
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2